Koa Shen

Mechanical Engineer
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EDUCATION

University of California, Santa Barbara — Expected June 2028
B.S. Mechanical Engineering | Major GPA: 3.97 | Cumulative GPA: 3.94 — Santa Barbara, CA

- Dean's Honors every quarter. Coursework: Dynamics, Circuits, MATLAB. Current: Dynamical Systems and Machine Learning. Planned: Mechatronics, Robotics Lab, Inverse Kinematics.

CONTROLS PROJECTS

Closed-Loop Cycloidal Actuator | Personal Project — Summer 2026 - Present

- Built a closed-loop stepper joint: an Arduino Uno commands a TMC2209 driver, a 12-bit AS5600 magnetic encoder measures motor-shaft position, and firmware detects position error and issues compensating moves.
- Wrote PlatformIO C++ firmware for phase-based bring-up, step/dir pulse generation, and real-time CSV telemetry of time, step position, and measured angle.
- Bench-validated automatic position-error correction and missed-step detection on a single-joint test setup.
- Designed a backlash test that preloads the shaft against a limit switch and uses encoder feedback to measure travel before the switch changes state.
- Designed a 15:1 cycloidal reducer within a NEMA 17 motor's 42 mm face; planned multi-joint upgrades add output-axis encoders, a TCA9548A I2C multiplexer, and Drake-based trajectory optimization.

EXPERIENCE

Mechanical Engineering Intern — Summer 2026 - Present
SLAC National Accelerator Laboratory, LCLS — Menlo Park, CA (part-time remote)

- Built an open-source Python/Drake kinematic model of a 30-DOF, 305-body assembly of linear, rotary, and tip-tilt motion stages; millisecond collision checks exposed at least five physical pinch points.
- Identified 10+ actionable design changes, including encoder retrofits and repositioned stage stacks.
- Developing EPICS controls compatibility and path planning, laying groundwork for homing sequences and safe/no-go motion zones; 15 reusable stage models are being adopted across three LCLS assemblies.
- Designed mounts and detector strain relief, checked stage and mount load cases for $100k+ detectors, and released Solid Edge drawings with GD&T through Teamcenter PDM.

Fabrication Team Lead — Fall 2024 - Present
Gaucho Racing, Formula SAE Electric | UC Santa Barbara — Santa Barbara, CA

- Helped develop a steering-wheel driver interface integrating buttons, potentiometers, a display, and a quick release; FEA-guided design reduced mass from 5 lb to 1.7 lb.
- Led 15 teammates through GR26 driver-safety-system design, fabrication, and installation; the car passed FSAE technical inspection and competed in every dynamic event.
- CNC-machined 33 steel and aluminum suspension parts, holding bearing fits within 0.0005 in.

Maintenance Technician — Summer 2025
Exploratorium — San Francisco, CA

- Maintained electromechanical exhibits, including an encoder-controlled spinning table and a piston-crank drive; led three major overhauls across 100+ serviced exhibits.
- Redesigned rotary-encoder mounts to resist garnet-sand ingress and simplify maintenance; manually machined replacements onsite.

TECHNICAL SKILLS

Controls and Embedded: Closed-loop position control, encoder feedback, stepper drivers (TMC2209), AS5600 magnetic encoders, I2C, Arduino, PlatformIO, data logging
Software and Modeling: C/C++, Python, MATLAB, Drake, kinematics, dynamics, collision detection, Git
Mechanical: SolidWorks, Solid Edge, Teamcenter PDM, GD&T, FEA, CNC/manual machining, 3D printing